

ABDUL MUQTADIR

Avionics, RF & Embedded Systems Engineer

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RESEARCH INTERESTS

RF & Microwave Engineering, 3D Full-Wave Electromagnetic Simulation, Super Wideband (SWB) & Conformal Traveling-Wave Antennas for Electronic Warfare (EW), Software-Defined Radio (SDR) Telemetry Systems, Closed-Loop Multivariable Digital Control (PID, State-Space, LQR), and Cyber-Physical Robotics.

EDUCATION

National University of Sciences and Technology (NUST)

Sep 2023 – Expected Jun 2027

Bachelor of Engineering in Avionics Engineering; CGPA: 3.73 / 4.00 (Semester 7)

College of Aeronautical Engineering, Risalpur

- **Academic Honors:** Dean's Honor List across all 7 consecutive semesters.
- **Core Coursework:** RF & Microwave Engineering, Antenna & Wave Propagation, Digital Signal Processing (DSP), Control Systems Engineering, Microcontrollers & Embedded Systems, Electronic Warfare, Radar Systems, Flight Telemetry.

The Universal Millennium College

Aug 2021 – May 2023

Cambridge International A-Levels — Mathematics (A), Physics (A), Chemistry (A)

Pakistan

Bloomfield Hall School

2008 – 2021

Cambridge International O-Levels — 1 A, 6 A Grades (Science & Mathematics Stream)*

Pakistan

TECHNICAL EXPERTISE & LABORATORY TOOLS

RF, Microwave & Antennas: CST Studio Suite 2025 (Time & Frequency Domain 3D Solvers), Keysight ADS, Keysight LineCalc, S-Parameter Characterization (S11 return loss, S21 insertion loss/isolation), Co-Planar Waveguide (CPW) & Microstrip Design, Vivaldi Antennas, Aerodynamic Conformal Arrays.

Communications & SDR: ADALM-Pluto Software-Defined Radio, 915 MHz FM Telemetry, Quadrature Demodulation, Digital IIR/FIR Filter Design, Additive White Gaussian Noise (AWGN) Filtering.

Control Systems & Dynamics: Closed-Loop Digital PID Control, Linear Quadratic Regulator (LQR), State-Space Modeling, IMU 6-DOF Sensor Fusion (Kalman & Complementary Filters), Lagrangian Equations of Motion, MATLAB / Simulink.

Programming & Firmware: C, C++, Embedded C, Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, G-Code, Linux/Bash, Git/GitHub.

Hardware & Instrumentation: ESP32 Dual-Core, STM32, Arduino Uno, MPU6050 IMU, BTS7960 43A Motor Drivers, Stepper Motors, Digital Storage Oscilloscopes (DSOs), Spectrum Analyzers, CATIA V5, AutoCAD, KiCAD.

RESEARCH & ENGINEERING PROJECTS

Super Wideband (3–18 GHz) Conformal Vivaldi Antenna for Electronic Warfare

Aug 2026

Lead Designer | CST Studio Suite 2025, Rogers RT6035HTC, CPW Transition

NUST CAE

- Designed and numerically simulated an ultra-compact planar and aerodynamically conformal Vivaldi antenna achieving a 6:1 impedance bandwidth ratio with return loss $S_{11} < -10$ dB across 3.0 to 18.0 GHz for airborne Electronic Support Measures (ESM).
- Synthesized a single-layer Co-Planar Waveguide (CPW) feed transition eliminating via inductance, and formulated an exponential flare profile (opening rate $R = 0.16 \text{ mm}^{-1}$) with 4 curved ground slits to suppress destructive higher-order cross-polarization modes.
- Modeled aerodynamic conformal cylindrical curvature matching UAV fuselage contours and validated inter-element mutual coupling isolation ($S_{21} < -25$ dB).

Hardware-in-the-Loop (HIL) UAV Altitude Control & SDR Telemetry

Jul 2026

Systems & Communications Engineer | ADALM-Pluto SDR, Python, PID Autopilot

NUST CAE

- Constructed a real-time HIL telemetry framework coupling a simulated second-order aircraft climb dynamics plant with a closed-loop digital PID autopilot.
- Modulated continuous altitude telemetry over a physical 915 MHz RF link using an ADALM-Pluto SDR with frequency modulation (FM).
- Constructed a receiver DSP pipeline performing quadrature demodulation and single-pole IIR noise filtering to reconstruct the autopilot feedback loop under simulated AWGN channel conditions.

2.4 GHz 5th-Order Chebyshev Stepped-Impedance Low-Pass Filter

Jun 2026

RF Designer | Keysight ADS, Keysight LineCalc, CST 3D EM, FR4 Microstrip

NUST CAE

- Synthesized a distributed stepped-impedance microstrip low-pass filter at a 2.4 GHz cutoff frequency for avionics ISM-band harmonic suppression.
- Executed Monte Carlo statistical yield analysis in Keysight ADS to model substrate permittivity variations (dielectric constant $\epsilon_r = 4.4 \pm 0.1$) and trace width tolerances; verified full 3D S-parameters in CST Studio Suite.

Two-Wheeled Inverted Pendulum Robot: LQR vs. Cascaded PID Analysis

May 2026

Control Systems Engineer | MATLAB/Simulink, Embedded C++, MPU6050 6-DOF IMU

NUST CAE

- Derived non-linear Lagrangian equations of motion and linearized state-space matrices around the unstable vertical equilibrium.
- Demonstrated physical degradation of theoretical state-feedback LQR under unobserved linear cart states (position and velocity); engineered and stabilized a physical prototype via cascaded PID firmware with anti-windup clamping and complementary sensor filtering.

Discrete Multi-Waveform Analog Function Generator

Mar 2026

Analog Circuit Designer | NE555 Core, Active Op-Amp Integrator & Shaper, Multisim

NUST CAE

- Designed and physically validated a discrete multi-waveform analog generator providing square, triangle, and sine waveforms from 10 Hz to 50 kHz.
- Cascaded active op-amp integrator and diode wave-shaping network producing smooth sinusoidal outputs with low total harmonic distortion (THD < 2.5%).

CNC 2D Handwriting Machine & Vector Plotter

Dec 2025

Mechatronics Designer | Arduino Uno, GRBL v1.1, A4988, NEMA 14, CATIA V5

Personal Project

- Designed dual-axis mechanical gantry in CATIA V5 and calibrated GRBL firmware EEPROM to achieve 0.05 mm positional repeatability for vector G-code execution.

Autonomous Heavy Competition Rover

Apr 2025

Hardware & Power Engineer | ESP32 Dual-Core, Dual BTS7960 43A, 4S2P Li-Ion

Robotics R&D

- Engineered heavy-duty powertrain utilizing dual BTS7960 43A H-bridges, JGA-25-370 geared motors, and custom 16.8V battery management with real-time wireless telemetry.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Freelance Engineering Consultant

Jul 2025 – Present

Independent Contractor — Embedded Systems, Firmware & RF

Remote

- Consult on microcontroller firmware architecture, I2C/SPI sensor interfaces, and closed-loop control algorithms for international engineering clients.
- Perform schematic reviews, circuit debugging, and component selection for high-reliability embedded hardware builds.

Head of Avionics Club

Sep 2024 – Present

College of Aeronautical Engineering, NUST

Risalpur, Pakistan

- Direct technical initiatives, SDR telemetry experiments, and autonomous robotics workshops for 40+ engineering students.
- Organize technical bootcamps bridging analytical classroom physics with hands-on hardware laboratory implementation.

A-Level STEM Tutor (Mathematics & Physics)

Nov 2022 – Present

Private Mentorship

Remote & In-Person

- Mentored 20+ students in differential calculus, kinematics, wave mechanics, and electromagnetism, with students achieving top A/A* grades.

HONORS & CERTIFICATIONS

- **Dean's Honor List:** Awarded across 7 consecutive semesters for top academic standing at NUST CAE.
- **IBM Professional Certification:** Data Analysis with Python (NumPy, SciPy, Pandas, Scikit-Learn).